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## SURVEY

# Systematic Reviews in the Engineering Literature: A Scoping Review

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**ABSTRACT** A systematic review is a specialized type of literature review used to collect and synthesize all the available evidence related to a research question. The methods for systematic reviews should be transparent and reproducible so that other researchers can use, replicate, and build upon the findings. Systematic reviews have been published for decades in medical literature where it is necessary to bring together all the studies related to a particular disease or condition for healthcare professionals to make evidence-based recommendations. Conducting systematic reviews has expanded to other disciplines, including education, business, and engineering. The authors of this study conducted a scoping review to investigate how prevalent systematic reviews are in engineering and to what extent engineering authors are following published reporting guidelines. To conduct this study, the authors searched the databases Compendex, Inspec, and ERIC and retrieved 11,588 records. After removing duplicates and applying inclusion and exclusion criteria, 3,066 articles remained in the final data set. The authors then used a checklist of bibliographic and quality-related items to extract data from each of these studies. The findings show that systematic reviews are a popular and rapidly growing methodology in engineering, internationally, but the quality of the studies, in terms of how authors construct and report their methods, is often low, resulting in many studies that are irreproducible and may fall short of the goal of gathering all the available evidence on a topic. The authors include recommendations for engineering authors, higher education administrators, and publishers.

**INDEX TERMS** Engineers, libraries, librarians, reporting guidelines, methods, reproducibility.

## I. INTRODUCTION

In academic publishing, review articles are a methodology where authors collect and present evidence from multiple studies to evaluate a research question. Within the review methodology, there are multiple types of reviews that encompass various methods for collecting and presenting the evidence. Grant and Booth [1] identified 14 distinct

review types and methodologies, ranging from narrative literature reviews to systematic reviews (SRs) and meta-analyses (MA). Many researchers are familiar with narrative literature reviews, defined as "...synthesis or description of the literature review without using quantitative methods. Often the purpose of the review involves the evaluation of some set of investigations and involves theoretical statements and casts a wide range of topics and investigations" [2]. With narrative literature reviews, there are no formal guidelines for collecting and presenting evidence as authors have the

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freedom to choose which studies they present in their review. Conversely, SRs call for more rigorous methods. The exact definition of SRs varies from source to source, but a representative definition is:

Systematic review describes a specific methodology for conducting reviews of literature. This methodology prescribes explicit, reproducible, and transparent processes for collating the best available evidence in answer to specific questions. In particular, it requires the use of robust techniques for searching for and identifying primary studies, appraising the quality of these studies, selecting the studies to be included in the review, extracting the data from the studies, and synthesizing the findings narratively and/or through pooling suitable quantitative data in META-ANALYSIS [3].

A 2019 study [4] proposed a new definition for SRs that prescribes clearer language focusing on the main components that should occur in an SR, based on the authors' analysis of the definitions used in 188 health science SR articles. They proposed that to be classified as an SR an article should include:

- 1) research question
- 2) sources that were searched, with a reproducible search strategy (naming of databases, naming of search platforms/engines, search date and complete search strategy)
- 3) inclusion and exclusion criteria
- 4) selection (screening) methods
- 5) critically appraises and reports the quality/risk of bias of the included studies
- 6) information about data analysis and synthesis that allows the reproducibility of the results.

As the methodology has become increasingly prevalent, the methods used throughout SRs have evolved as well, including the use of artificial intelligence or machine learning in various stages of the review process, as evidenced in the 2020 PRISMA guidelines [5]. The importance placed on well designed and reported SRs is further evidenced by the detailed instructions provided by the accompanying PRISMA extension articles for searching [6] and SR and MA protocols [7].

The benefits of conducting SRs are that they include and assess all the relevant evidence related to a research question. The drawbacks to SRs are that they are only as good as the methods used to discover and evaluate the existing research, and their success is reliant upon the quality of the evidence researchers are able to access [8]. To address these concerns several organizations have developed guidelines and recommendations for applying the SR methodology in multiple disciplines, however, not all researchers are aware of or adhere to existing guidelines [9], [10], [11]. As the popularity of SRs has grown, so have disciplinary focused articles on the conducting and reporting of SRs with examples in engineering education [12], software engineering [13],

[14], conservation and environmental management [15], and systems and automatic engineering [16].

Given the importance often associated with the use of SRs in decision making and the rise in published SRs, previous studies have investigated the prevalence and quality, to varying degrees of depth, of published SRs in specific disciplines including conservation biology [17], software engineering [18], [19], crop science [20], and clinical neurology [21].

These studies have focused on specific disciplines. To the best of our knowledge, only one other study has looked broadly at engineering [22]. Our study expands on this previous research study through a more inclusive search strategy and an investigation into how engineering researchers described the methods used in their SRs or MAs. This study builds on our previously reported work-in-progress paper [23] and our published subset analysis focused on engineering education [24]. SRs can also be referred to as systematic literature reviews, but for the purposes of this study, we will refer to all included records as systematic reviews (SRs). Additionally, for this study, MAs that acquired their data through a literature search of data were included and are also referred to as SRs, except in particular methods and results sections where it is important for the reader's understanding to include the term MA. Lastly, we conducted this study as a scoping review, as opposed to a systematic review, because our goal was to provide an overview of a large body of literature on a broad topic, rather than to provide an answer to a narrowly focused research question.

The research questions for this scoping review are:

- RQ1. To what extent is the systematic review (SR) research method being applied in engineering disciplines?
- RQ2. How closely are SRs that are published in engineering disciplines following established guidelines?

These questions were investigated across the broad engineering disciplines with selected components presented on a sub-disciplinary level (e.g., industrial, mechanical), as defined by the National Academy of Engineering [25].

## II. METHODS

This review was conducted according to best practices in scoping reviews and is reported according to the PRISMA scoping review reporting guidelines [5], [26]. The methods for this project have been previously published in the Journal of Engineering Education, reporting a sub-analysis of this study focused on engineering education [24].

### A. PROTOCOL AND REGISTRATION

A protocol was created a priori and uploaded to Open Science Framework (OSF) (available at <https://osf.io/de3mq/>), an open repository frequently used for review protocols because it creates timestamps of uploads and edits and is freely accessible.

## B. ELIGIBILITY CRITERIA

### 1) INCLUSION CRITERIA

To qualify for inclusion in this study, documents must have satisfied the following conditions:

- 1) The documents must be journal articles or conference papers. We chose to include conference papers because it is common practice in engineering disciplines to present peer-reviewed work at conferences.
- 2) The full text must be in English. Given our language skills, we are not able to fully assess documents in other languages.
- 3) The study type must be a systematic review and/or a meta-analysis that includes an SR-style search, or the authors intended to follow one of those methodologies. We excluded studies that claimed to be SRs or MAs but shared little or no details (e.g., database selection, search terms) on their data collection methods. This choice was made as it was unclear if the authors were trying to use these methodologies or if they coincidentally used those terms.
- 4) The document must include a focus on engineering, excluding software engineering and computer science. We define “a focus on engineering” as (A) an author with an engineering affiliation, or (B) publication in a journal or conference in the engineering field, or (C) the application of engineering expertise in non-engineering subject areas (e.g., medicine) or (D) for engineering education studies, if engineering or STEM or STEAM appeared in the article title. For studies with STEM or STEAM in the article title, studies were only included if there was an engineering education component.

### 2) EXCLUSION CRITERIA

- 1) Studies that were not journal articles or conference papers were excluded, such as book chapters or dissertations. We chose not to include other document types because they are not typically used to disseminate SRs and MAs.
- 2) Studies where the full text was not available in English were excluded.
- 3) Studies that did not identify as a systematic review, systematic literature review, or meta-analysis in the title or abstract were excluded. Studies were also excluded that claimed to be SRs or MAs but did not report anything in their methods that allowed us to extract data.
- 4) Studies that were identified as works in progress or preprints were excluded. As these works have not completed the publishing process, we believed there was potential the final methods reported might change based on completing the project and going through peer review.
- 5) Studies that did not include a focus on engineering were excluded, as were studies that were software engineering or computer science. We chose not to

include software engineering and computer science because they have a distinct and long-standing set of disciplinary guidelines for SRs already in place [27]. We did include engineering education related studies that had aspects of software engineering (e.g., software engineering students). For articles with STEM or STEAM in the article title, studies were excluded if the study did not include engineering education, for example, STEM in agriculture education.

## C. INFORMATION SOURCES

Compendex via Engineering Village platform (coverage from 1884), Inspec via Engineering Village platform (coverage from 1896), and Education Resources Information Center (ERIC) via EbscoHost platform (coverage from 1966) were searched.

## D. SEARCH

The search strategy was created by an engineering librarian (MP) with experience working on evidence synthesis projects and was reviewed by two other librarians, one with experience working on SRs in engineering (ASVE) and one with experience working on evidence synthesis projects in the health sciences (JBR). The original search was created in Compendex and then translated to other databases. The search combines keywords for SRs with engineering disciplines as defined by the National Academy of Engineering subject list [25]. The keywords remained constant across all databases, with subject thesaurus terms added to the ERIC search. For Compendex and Inspec, preprints were removed from the results using NOT and the preprint document type. This step was not conducted in ERIC because preprints are not as prevalent in that database. The original search was run on July 25, 2019, and was updated on October 24, 2022. The search in ERIC was only performed on the second date, however, no date limit was applied to that search in order to capture the full range of potential results. See Fig. 1 for the full search in Compendex and Inspec and Fig. 2 for the full search in ERIC.

## E. SELECTION OF SOURCES OF EVIDENCE

For the original search, duplicates were identified manually using a validated method in EndNote X9 [28]. After duplicates were removed, the remaining results were uploaded into Rayyan [29] for the title/abstract review. Three authors (MP, JBR, ASVE) conducted an initial pilot test of the inclusion and exclusion criteria by reviewing a random selection of 100 titles and abstracts. The reviewers had an 88.7 percent agreement rate with a Fleiss kappa inter-rater reliability percentage of 43.7 percent (moderate agreement). All disagreements were discussed by all three authors to develop a consensus. After pilot testing was completed, the remaining records were divided equally among the three authors for single review. To reduce the potential for excluding relevant articles due to single review, an inclusive

```
(((((("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis") WN
KY) AND ((("aerospace engineering" OR "aeronautical engineering" OR "astronautical engineering"
OR "bio* engineering" or "biomedical engineering" OR "chemical engineering" OR "civil
engineering" OR "construction engineering" or "environmental engineering" or "structural
engineering" or "electronic engineering" OR "electrical engineering" OR "computer engineering" OR
"energy engineering" OR "nuclear engineering" OR "industrial engineering" or "manufacturing
engineering" or "systems engineering" OR "operations research" OR "materials engineering" OR
"mechanical engineering" OR "engineering education") WN ALL)))) NOT ((pp) WN DT))
```

**FIGURE 1.** Full search strategy for compendex and Inspec.

1. DE "meta-analysis" OR DE "State of the Art Reviews"
2. TI ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis") OR AB ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis") OR KW ("systematic review" or "systematic literature review" or "meta-analysis" or "meta analysis")
3. S1 OR S2
4. DE "Engineering Education" OR DE "Aerospace Education" OR DE "Engineering" OR DE "Agricultural Engineering" OR DE "Chemical Engineering" OR DE "Civil Engineering" OR DE "Human Factors Engineering"
5. TI engineering OR AB engineering OR KW engineering
6. S4 OR S5
7. S3 AND S6

**FIGURE 2.** Full search strategy for ERIC.

approach was made where results were only excluded if they clearly met one of the exclusion criteria based on the title or abstract. Once completed, the full-text articles were retrieved and uploaded into Rayyan. At this stage, a fourth author (DZ) was recruited to help with full-text screening. Two rounds of pilot testing, consisting of a random selection of 44 full texts in each round, were conducted to ensure agreement among the reviewers in applying the inclusion and exclusion criteria at this stage. A common discussion during the pilot testing involved articles that included terms for SRs or MAs in their title or abstract but did not follow reporting guidelines for those methodologies. Our group determined that if a study self-identified as an SR and/or MA and the authors reported in their methods at least one category we could extract, that study would be included. This may have led to the inclusion of studies that were not intended by the authors to be one of these methodologies and may have led to the exclusion of studies that were meant to be SRs but did not report any of the elements we were evaluating. Based on these discussions and in keeping with best practices, two reviewers were assigned to independently review each full text. In cases of disagreement, the two reviewers discussed the article together, and in cases where consensus could not be met, the article was discussed by the entire group. For the updated search duplicates between databases and from the original search were removed using the same process as the original search. The results from the updated search were uploaded into Covidence [30] for screening. The screening process followed the same steps as the original search, except the fourth author (DZ) was added to the title/abstract step as well for the updated search.

#### F. DATA EXTRACTION PROCESS

Data was extracted from each included article using Google Sheets. The data categories were drawn from the PRISMA checklist [31] and from two evidence synthesis quality checklists, AMSTAR2 (A Measurement Tool to Assess systematic Reviews) [32] and Share+DART (Documentation and Appraisal Review Tool) [33], to create a custom form for this project that focused on the methods used and reported. For the original search, a pilot test of the data extraction sheet was conducted by having four authors (MP, JBR, DZ, ASVE) extract data from a random selection of 18 articles. The four authors compared the results, any discrepancies were discussed, and clarifications were made to the extraction sheet as needed. After the pilot test, the remaining records to be extracted were equally divided among four authors for solo extraction. Regular check-in meetings were scheduled and each reviewer could bring questions to the group for discussion. Areas of frequent discussion included discipline, due to the interdisciplinary nature of engineering research and discussions on assessing risk of bias, where studies did not include a specific tool for measuring quality/risk of bias.

For the updated search, given the large number of studies found and an interest in being able to present the findings in a timely manner, an additional five authors (AGB, EMR, QMZ, JMC, WZ) were recruited to assist with data extraction, while one previous reviewer (ASVE) had to step away to focus on other projects. To onboard the new extractors, an initial meeting was scheduled to walk through the data extraction sheet with an explanation for each category. Following this meeting, all extractors (MP, JBR, DZ, AGB, EMR, QMZ, JMC, WZ) participated in a new round of pilot testing



involving five records and another meeting to discuss any questions and discrepancies that occurred. Upon completion of the pilot testing discussion, the items to be extracted were divided equally among the eight extractors for solo extraction, with each person extracting approximately 220 records. Monthly meetings were scheduled so that questions about specific data extractions could be brought to the group for discussion and the notes from each session were shared through a group folder. This allowed all team members to review the document during the data extraction process, again to ensure consistency across responses.

### G. DATA ITEMS

In an effort to ensure accuracy, the text describing the data items is repeated from our previously published article that focused on the subgroup of engineering education, with some edits for clarity [24]. One update from that paper was not to extract data on whether there was an explicit statement on whether the methods were established before completing the SR because that category was deemed redundant to having registered a protocol after reviewing the initial data. The data extraction sheet and the full data extracted can be found in the OSF project for this review (<https://osf.io/de3mq>). To help with consistency across reviewers, most data columns included predefined drop-down options, exceptions included basic article information and the list of sources searched. After the pilot round of data extraction, a table of naming conventions for databases and source titles was created to help create more consistency in these columns for the second round of data extraction.

### 1) BIBLIOGRAPHIC DATA

- **Basic Article Information** For each included article we extracted the last name for the first author, the year the article was published, the country of the corresponding author (or the first author if the corresponding author was not reported), the source title (journal or conference proceeding where the article was published), and the publication title (title of the article or conference paper).
- **Engineering Discipline** The list of engineering disciplines was the same list, based on the National Academy of Engineering subject list [25], used in creating the search strategy and in the criteria for inclusion in the final data set. This data element did involve a level of subjectivity that required a judgment call because of the amount of multidisciplinary work performed within engineering research. When multiple disciplines were represented in an individual article, we checked the journal or conference title to see if a specific discipline was present and checked the first or corresponding author's affiliation. If a specific engineering discipline was not listed in these sources, we went down the author list and selected the first discipline listed in an author affiliation.
- **Systematic Review (SR) or Meta-analysis (MA) in Title** The next two data extraction elements were if

the final article self-reported as an SR and/or MA in the title and if so which one. A drop-down option for yes or no was used for reporting in the title. For which variation, the drop-down options were SR, MA, both, neither. SR was selected for titles that included systematic review or systematic literature review and MA for titles that included meta-analysis but not SR.

### 2) REVIEW PROTOCOL DATA

- **Clearly Defined Research Question** For this category we looked for an explicit research question, often located at the end of the introduction. To be marked as included the authors needed to explicitly state it as the research question, variations like Q1, or in question format. Studies that reported the objective of the study as performing an SR of the literature on the topic of the paper were marked as no for this criterion. We made this decision as one of the purposes of an SR is to answer a specific research question and when the data extraction sheet was created the PRISMA reporting guidelines specifically include a research question. During this study, PRISMA updated their reporting guidelines to include research objectives as an option, however we kept this criterion as initially defined, for the purpose of having one standard metric across all included studies.
- **Registered Protocol** Registering an SR protocol promotes transparency and accountability for sticking to the a priori nature of conducting an SR. While most SRs in engineering outside of biomedical engineering may not be eligible for registration with PROSPERO, a registry used for systematic reviews with health outcomes, researchers could register protocols with repositories such as Open Science Framework (OSF). Authors had to explicitly state where their protocol was registered or provide information on how to access their protocol.
- **Explicit Statement of Inclusion and Exclusion Criteria** To be marked as yes for this component, authors had to list the predetermined criteria that they used to determine the eligibility of discovered studies in the final SR. Criteria could be presented as either a figure or described in text but needed to be explicitly stated as being inclusion or exclusion criteria.
- **PRISMA Flowchart or List of Excluded Studies and Reasons Why** The PRISMA flowchart is a well-known and often used figure that shows the progress of studies in SRs from the initial search to final inclusion [5]. The purpose of this flowchart is to provide transparency on how the review criteria was applied in the study. Any study that included either the official PRISMA flowchart or their own flow diagram showing the number of studies included and excluded at each stage of the review was marked as yes for this category. Additionally, if study authors described in text the flow of articles in each step of the review and why articles were excluded then those articles were marked yes as well.

- **Explicit Statement of Assessing Risk of Bias of Included Studies** To receive a yes for this criterion studies needed to either explicitly state that articles included in the SR were assessed for quality or risk of bias or the authors needed to include the name of the tool used to assess quality of each included study. Statements along the lines that because only peer reviewed articles were included or that only quality databases were searched meant all included studies were of high quality were marked as no.
- **Explicit Statement of Multiple Reviewers or Explanation of Interrater Reliability** In order to be classified as yes for this criterion, studies needed to explicitly state that each record was reviewed for inclusion in the SR by more than one reviewer, or they needed to describe their process for piloting the process or provide an interrater reliability measure. While interrater reliability is not a perfect measurement, it demonstrates that study authors were aware of the importance of verifying that different authors might apply criteria differently.

### 3) SEARCH DATA

- **Date Search Run** In order to be marked as yes for this category, articles needed to provide at least the month and year that the searches were run. Providing this information is important for the reproducibility of SRs as it provides additional context for what articles the researchers could have discovered with their search. We did not require the reporting of the specific day, as many engineering related databases do not provide a search filter option for a specific day of indexing in the database.
- **At Least One Full Database Search String Provided** In order to be marked as yes for this category authors needed to provide either a figure of the search from one of the databases or provide enough detail in the written description to allow a reader to replicate the search in the reported database. This would include the specific keywords used, the search fields used, and how the terms were combined.
- **Database/s Search Strategy Provided (General Themes or Concepts)** To be marked as yes in this category authors needed to provide a list of terms or concepts they used in their search for articles, but not necessarily the level of detail needed to reproduce the search. All studies marked as yes for reporting at least one full database search string were also marked yes for this category.
- **Search Filters Identified** To be marked yes in this category, authors had to report any limiters or search filters applied to their search, this could be reported in either the full search string or as text detailing the search process. Studies were also marked yes if they explicitly stated no search filters or limits were applied. Examples of search filters that might be available and used for a SR include publication date, language of

article, and publication type (e.g., article, conference paper, abstract).

- **Search Quality** Studies were marked as high or low search quality based on all the factors related to the description of the search strategy. Studies could receive a yes for at least one full database search string provided but receive a low quality if the reported search was limited in scope, a single source searched or only a couple of keywords used. Each reviewer drew upon their experience as a database expert to make this determination.

### 4) SOURCES DATA

- **List of Sources** This was a yes or no category measuring if study authors reported where they searched for articles for their study. Sources could include databases, platforms (such as EBSCO or specific publishers like Wiley), repositories, or the specific journals searched.
- **Database Platforms Listed** An important component in reporting the search strategy is not just listing the sources searched, but also in the case of databases, which platform the database is hosted on. This is important as different platforms provide different search options and it can be impossible to replicate a search if the platform is not identified. On the reverse side is the reporting of what databases were searched if platforms are reported. For instance, EBSCO is a common research platform subscribed to by academic libraries, however, each library chooses which databases hosted by EBSCO they want to subscribe to and researchers can choose which of the databases available to them that they want to search. If a study reports searching EBSCO that is not enough information to determine whether appropriate databases were searched. To be marked as yes for this category, a study had to report the database and platform for each source that is available in multiple platforms.
- **Total Number of Databases / Platforms / Repositories Searched** This element is a numerical count of the total number of databases / platforms / repositories listed. If authors reported searching a platform that hosts multiple databases, but did not specify which databases they searched, that was counted as one. If Google Scholar was reported in the list of other databases searched that was included in the count. However, if Google Scholar was mentioned as an additional component of the search that was not included in this count. This decision was made as some researchers use Google Scholar as part of their strategy for finding unpublished reports or tracking citing or cited articles and not as part of the reproducible search strategy. Specific journals that were searched were also not included in this count.
- **List of Databases / Platforms / Repositories Searched** Each of the sources that authors reported as being part of their search, excluding specific journals, were listed

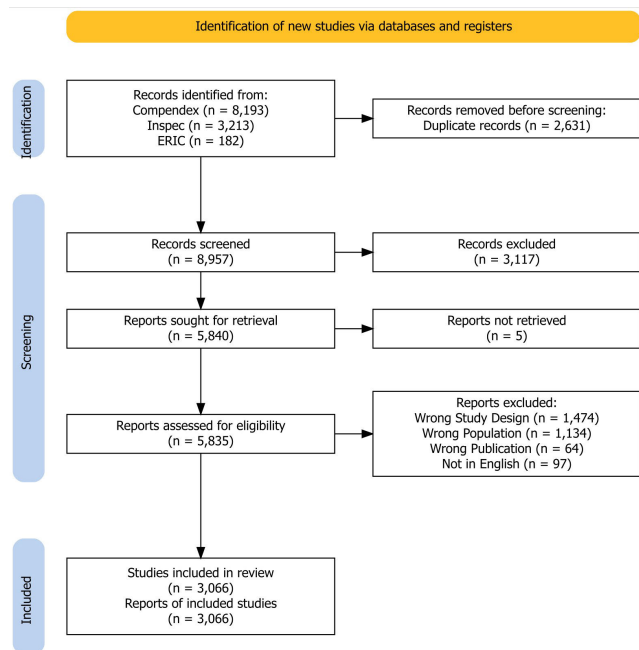


FIGURE 3. PRISMA flow diagram for this study.

here to investigate what sources are commonly used in engineering SRs.

#### 5) LIBRARIAN PARTICIPATION DATA

- **Librarian or Database Expert Listed as Author** Affiliations were checked for librarian or information expert identification.
- **Librarian or Database Expert Consulted** The full text was examined to see if the study authors reported meeting or discussing their search with a librarian or database expert. If included, this was listed in the methods or acknowledgment sections. Any studies marked as yes for librarian or database expert listed as an author were also marked yes here.

#### H. SYNTHESIS OF RESULTS

After completing the data extraction process, the data captured in the individual Google Sheets spreadsheets was combined and cleaned by a team member (DZ). The cleaned spreadsheet was used to create the results detailed in the Results section. The data analysis was conducted using Google Sheets (MP) and R (QMZ).

### III. RESULTS

Fig. 3 details the flow of SRs through the identification and screening process for this scoping review. Overall, we gathered 11,588 SRs from searching the databases Compendex (Engineering Village), Inspec (Engineering Village), and ERIC (EbscoHost). After removing 2,631 duplicates, 8,957 SRs were screened at the title and abstract level. After applying the inclusion and exclusion criteria in a title and abstract level review, 3,117 records were excluded and 5,835 SRs remained for the full text level review. At the

TABLE 1. SR count percentage change by year, 2015-2022.

| Time Period | Percentage change of SR counts |
|-------------|--------------------------------|
| 2015-2016   | +36                            |
| 2016-2017   | +33                            |
| 2017-2018   | +56                            |
| 2018-2019   | +25                            |
| 2019-2020   | +19                            |
| 2020-2021   | +48                            |
| 2021-2022   | -2                             |

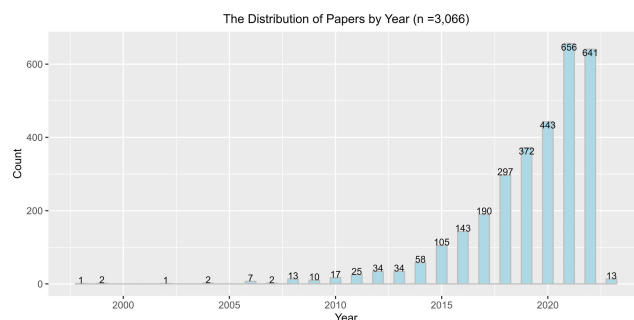


FIGURE 4. The distribution of papers in this scoping review by year.

conclusion of the full text level review, 3,066 SRs remained, making up the final dataset. A spreadsheet containing the metadata of the studies in the final dataset and each article's data extraction details can be found in our OSF Project (<https://osf.io/de3mq/>).

#### A. SUMMARY OF INCLUDED STUDIES

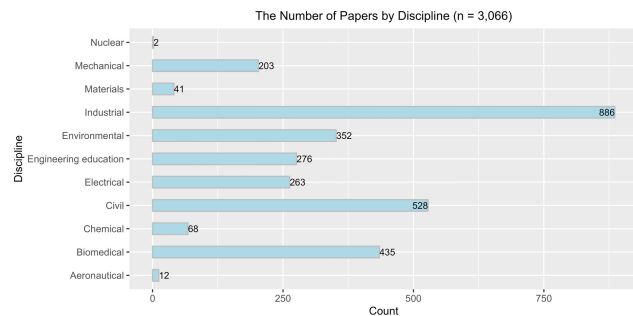
##### 1) DISTRIBUTION BY YEAR

While we put no time limitation on our search, the years of publication of the SRs in this scoping review range from 1998 to 2023. Few 2023 papers (13) appear in our data since our search was last run in October 2022, indicating the 2023 SRs that did appear were articles published as early access. Fig. 4 shows the distribution of SRs by year. Of the 3,066 included SRs, 93% were published from 2015-2023, showing the publication of SRs in engineering disciplines is a recent development. Table 1 shows the percentage increases year over year from 2015-2022. The largest increases occurred from 2017 to 2018 (56%) and from 2020 to 2021 (48%).

##### 2) ENGINEERING DISCIPLINES

Fig. 5 shows the number of SRs by engineering discipline and Fig. 6 shows the number of papers by engineering discipline by year. The five engineering disciplines with the highest counts were industrial (29%), civil (17%), biomedical (14%), environmental (11%), and engineering education (9%). The disciplines with the lowest counts were aeronautical (0.3%) and nuclear (0.07%). It should be noted that many of the SRs are multi-disciplinary, but we made efforts to identify a primary discipline by reviewing the publication title and corresponding or lead author affiliations.

If we contrast the publication numbers in Fig. 5 and Fig. 6 with data on the number of tenured or tenure-track faculty in accredited engineering departments at institutions



**FIGURE 5.** The distribution of papers in this scoping review by engineering discipline.

in the United States and Canada from the American Society for Engineering Education (ASEE) [34], we can see that the largest disciplines are not necessarily producing the most SRs. While this is not a direct measure of scholarly productivity and clearly does not account for every faculty member or researcher active in their fields, we can use it to make rough comparisons. According to ASEE, mechanical engineering constituted 17.1% of the total tenured or tenure-track engineering faculty in 2022, more than any other discipline, but mechanical engineering only accounts for 6.6% of the SRs in our data set. Chemical engineering, with 7% of the tenured or tenure-track engineering faculty, comprises 2.2% of the reviews in this study. Contrastingly, ASEE determined that industrial engineering accounts for 3.9% of the tenured or tenure-track engineering faculty, but they produced the most SRs in our data set with 28.9% of the total.

### 3) GEOGRAPHIC DISTRIBUTION

Fig. 7 illustrates the geographic distribution of SRs in this review, as determined by either corresponding or first authorship (see Methods). Overall, the 3,066 articles in the final dataset came from 89 different countries, despite our study being limited to articles written in English. The ten countries with the highest SR counts include the United States of America (413), Brazil (345), China (297), United Kingdom (176), Malaysia (151), Italy (147), India (138), Australia (113), Portugal (97) and Germany (90). The geographic distribution of our final dataset over time can be viewed at this link: <https://tinyurl.com/msae34nc>. The earliest SRs in the dataset (pre-2006) were published in Western European countries (e.g., Italy, United Kingdom), the United States, and Canada. The trend then expanded to Brazil (2006) and China (2008) and ultimately has spread to 89 different countries.

### 4) SOURCE TITLES

The 3,066 SRs are published in 1,029 different publication titles. Table 2 shows the twenty most frequent source titles in the dataset.

In publishing terms, we can see that SRs are being written for both scholarly journals and conferences. While the majority of the top source titles are journals, the ASEE Annual Conference and Exposition appears on the list at #7, with

**TABLE 2.** Most frequent source titles of the SRs in the engineering literature.

| Journal or Conference                                                                | Publisher        | SR Count |
|--------------------------------------------------------------------------------------|------------------|----------|
| 1. Journal of Cleaner Production                                                     | Elsevier         | 147      |
| 2. IEEE Access                                                                       | IEEE             | 62       |
| 3. Sensors                                                                           | MDPI             | 60       |
| 4. Computers & Industrial Engineering                                                | Elsevier         | 52       |
| 5. Renewable & Sustainable Energy Reviews                                            | Elsevier         | 52       |
| 6. Science of the Total Environment                                                  | Elsevier         | 48       |
| 7. ASEE Annual Conference & Exposition, Conference Proceedings                       | ASEE             | 40       |
| 8. Energies                                                                          | MDPI             | 33       |
| 9. International Journal of Production Research                                      | Taylor & Francis | 30       |
| 10. Sustainability                                                                   | MDPI             | 30       |
| 11. Automation in Construction                                                       | Elsevier         | 27       |
| 12. Engineering, Construction & Architectural Management                             | Emerald          | 27       |
| 13. IEEE International Conference on Industrial Engineering & Engineering Management | IEEE             | 24       |
| 14. IEEE Frontiers in Education (FIE) Conference                                     | IEEE             | 23       |
| 15. Safety Science                                                                   | Elsevier         | 23       |
| 16. Applied Sciences                                                                 | MDPI             | 19       |
| 17. Journal of Construction Engineering & Management                                 | ASCE             | 19       |
| 18. Production Planning & Control                                                    | Taylor & Francis | 19       |
| 19. Building & Environment                                                           | Elsevier         | 18       |
| 20. Journal of Engineering Education                                                 | ASEE (Wiley)     | 17       |

**TABLE 3.** Most frequent databases / repositories / platforms reported as searched by authors of engineering SRs.

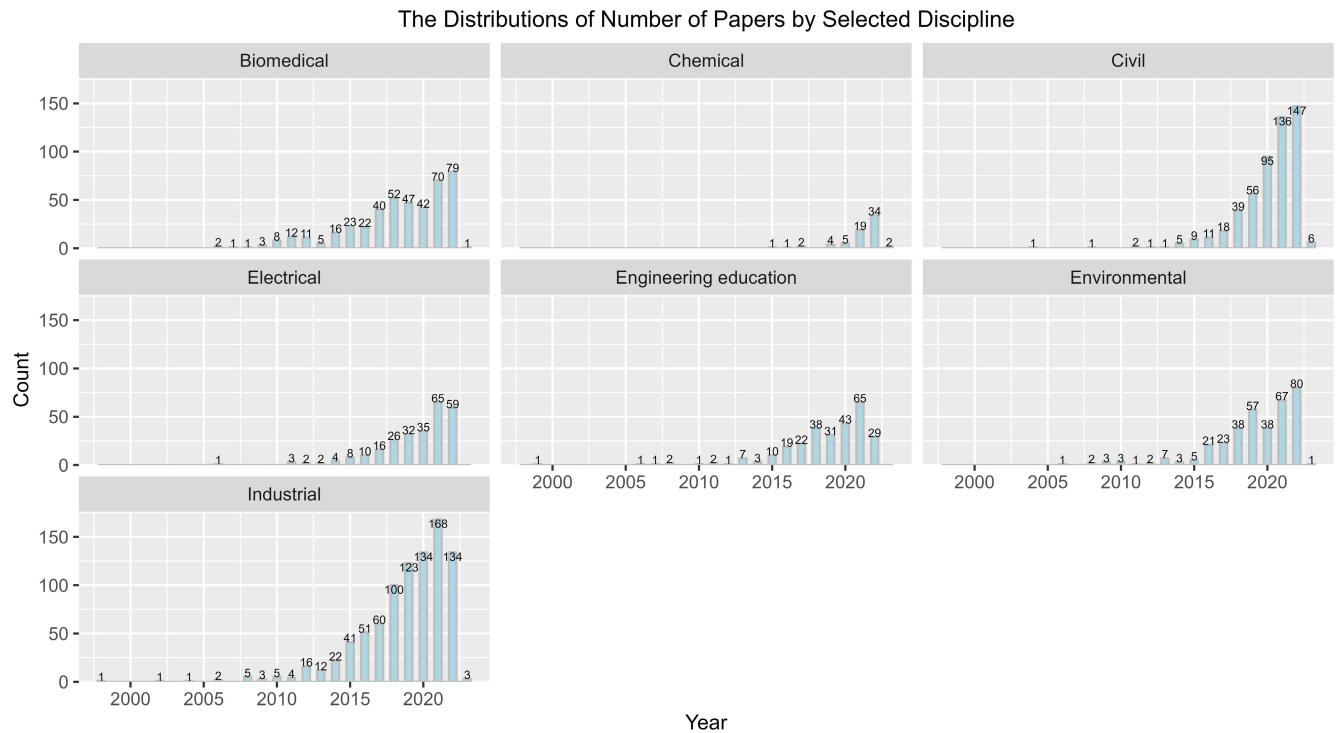
| Database / Repository / Platform | Article Count |
|----------------------------------|---------------|
| Scopus                           | 1,654         |
| Web of Science                   | 1,490         |
| ScienceDirect                    | 1,003         |
| Google Scholar                   | 707           |
| IEEE Xplore                      | 623           |
| PubMed                           | 528           |
| SpringerLink                     | 379           |
| Emerald Insight                  | 243           |
| ACM Digital Library              | 236           |
| MEDLINE                          | 200           |
| Wiley Online Library             | 179           |
| Taylor & Francis                 | 175           |
| ProQuest                         | 168           |
| EBSCOhost                        | 167           |
| EMBASE                           | 153           |
| Compendex                        | 105           |
| Engineering Village              | 103           |

IEEE International Conference on Industrial Engineering and Engineering Management and IEEE Frontiers in Education at #13 and #14, respectively.

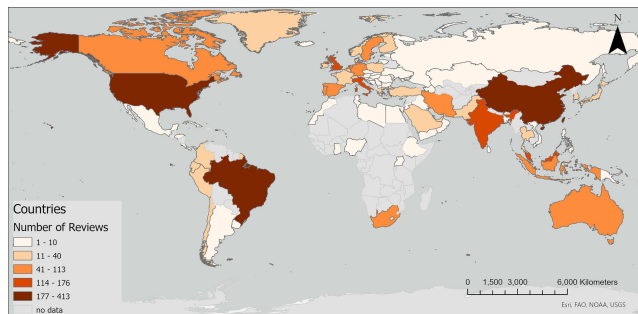
### 5) DATABASES / REPOSITORIES / PLATFORMS SEARCHED

In total, the authors of the 3,066 SRs reported they conducted literature searches in 518 distinct databases / repositories / platforms. As noted in the Methods section, this list does not include individual journal titles, even though some authors reported restricting their searches to a predetermined list of journals. Table 3 lists the names of the databases / repositories





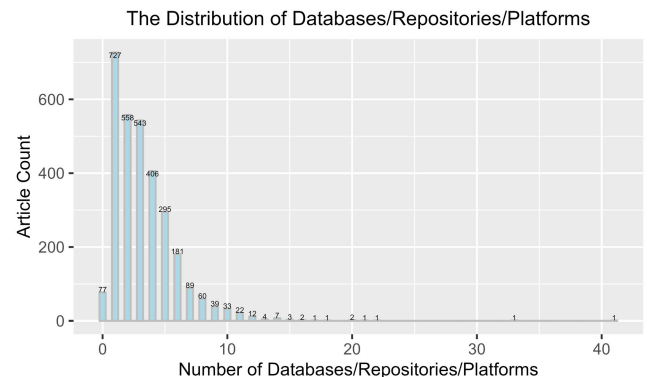
**FIGURE 6.** Count of papers by selected discipline by year. Aeronautical, materials, and nuclear engineering are not included due to low counts.



**FIGURE 7.** Geographic distribution of SRs in the engineering literature.

/ platforms that were reported in at least 100 different SRs. The most searched databases / repositories / platforms, each reported in more than 1,000 SRs, were Scopus, Web of Science, and ScienceDirect.

As discussed in previous papers that have analyzed SR reporting [24], [35], [36], it should be noted that because this list includes a mix of databases, repositories, and platforms, it is not possible to know the actual number and names of the databases searched in many of the SRs. The reason for this issue is that, with platforms such as EBSCOhost, Engineering Village, and ProQuest, libraries have options for choosing the individual databases they subscribe to on each platform, and different libraries may choose different options, due to cost, usability, etc. For example, when looking at the platform Engineering Village, libraries currently have the option of subscribing to Compendex, Inspec, and nine other databases [37]. If an author just lists “Engineering Village” as the resource they searched for literature, it is not possible

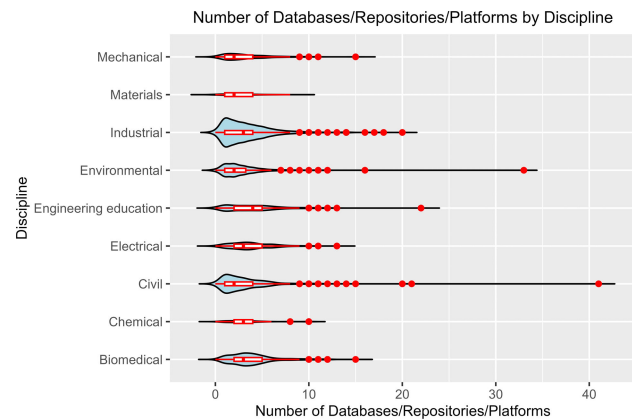


**FIGURE 8.** Frequency of the databases / repositories / platforms searched in the SRs ( $n=3,066$ ) included in this study.

for others to be able to replicate the search (one of the aims of SRs), because it is not clear which databases on the platform they actually searched.

Fig. 8 shows the frequency of databases / repositories / platforms searched. The number of databases / repositories / platforms reported by authors ranged from zero in 77 SRs to 41 in one SR. More than 25% of the SRs either did not list the names of any databases / repositories / platforms that were searched (77), or only described searching one database / repository / platform (727). The mean, median, and mode for the number of databases / repositories / platforms reported as searched were 3.3, 3, and 1, respectively.

Fig. 9 shows violin plots of the distribution of the number of databases / repositories / platforms reported by engineering discipline. The plots do not include nuclear engineering and



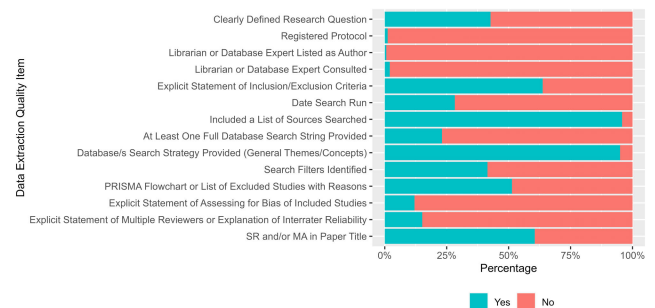
**FIGURE 9.** Violin plots of the distribution of the number of databases / repositories / platforms reported in SRs by engineering discipline. Nuclear and aeronautical engineering are not included due to very low counts.

aeronautical engineering since they contain a very small number of papers. The areas in blue are scaled proportionally to the number of papers, with the thicker areas representing a higher probability that papers will utilize the particular number of databases / repositories / platforms. The boxes in the center show the inter-quartile ranges and include markers indicating the median points in the discipline data. The median of each engineering discipline is between two and four. At four databases, engineering education has the highest median. The red dots show outliers. The farthest outlier is a civil engineering SR that reported searching 41 different databases / repositories / platforms.

## B. QUALITY ADHERENCE OF INCLUDED STUDIES

In this subsection, we share the results of the remaining data items we extracted from the 3,066 articles. Table 4 and Fig. 10 provide a summary of the full dataset with regard to the review protocol data items (see Methods), and the bibliographic data item focused on whether or not the authors included SR and/or MA in the paper title. The areas the SRs collectively performed the best were including a list of sources searched (96% Yes), providing at least the general concepts of a search strategy (95% Yes), including an explicit statement of inclusion and exclusion criteria (64% Yes), including “systematic literature review” or “systematic review” and/or “meta-analysis” in the title (61% Yes), and including a PRISMA flow chart or list of excluded studies with reasons (51% Yes). While 96% of the articles did include some type of list of the sources searched (i.e., databases, repositories, platforms, specific journals), this may not mean that readers would be able to precisely identify the sources searched. As previously mentioned, just listing a platform is not enough information to communicate the specific database(s) searched.

The areas where the SRs collectively performed the worst were including a librarian or database expert as an author (99% No), registering a protocol (99% No), consulting with a librarian or database expert (98% No), assessing the included studies for bias (88% No), and having an explicit statement



**FIGURE 10.** Percentage of SRs that adhered to (“yes”) or did not adhere to (“no”) each data extraction quality item (n=3,066).

of multiple reviewers or explanation of inter-rater reliability for the data screening process (85% No).

When we separated the overall number of studies into two subsets based on whether SR and/or MA appears in the paper title (see Table 5 and Fig. 11), the areas the SRs and MAs with the review type in title performed greater than those without review type in title were including an explicit statement of inclusion/exclusion criteria (71% Yes versus 52% Yes), and including a PRISMA flow chart or list of excluded studies with reasons (61% Yes versus 38% Yes).

Fig. 12 illustrates the paper counts by high or low search quality and whether a librarian was reported as being consulted with by the authors. The results show 53% of the papers that reported consulting with a librarian had a search considered high quality, whereas only 8% of the papers who did not report consulting with a librarian had a search considered high quality.

Tables 6 and 7 provide a geographical look at the metrics “Librarian or Database Expert Listed as Author,” or a “Librarian or Database Expert Consulted.” Of the SRs that reported consulting with a librarian, 45% were from the United States, 11% from Canada, 6% from China, 6% from Netherlands, and the remaining SRs from 13 other countries. Likewise, of the SRs that had a librarian co-author, 42% were from the United States, 16% from Netherlands, 11% from Canada, and the remaining SRs from six other countries.

## IV. DISCUSSION

Systematic reviews (SRs) are a popular and growing methodology in engineering disciplines, but the quality of the resultant articles and conference papers, in terms of how authors conduct their searches and report their methods, is often low. In this section, we will revisit our research questions and connect them to the data, make recommendations for improvement of SR quality and reproducibility, and suggest directions for future work.

### A. RQ1: TO WHAT EXTENT IS THE SYSTEMATIC REVIEW RESEARCH METHOD BEING APPLIED IN ENGINEERING?

From the data, it is clear the SR methodology is increasingly being applied in engineering disciplines. With Fig. 4 and Table 1, we can see the rapid rise in SR production over the course of the 2000s and 2010s, with an especially large

**TABLE 4.** Overall totals of engineering SRs adhering to data extraction quality items by SR phase (n=3,066).

| Data extraction quality item                                                      | Yes counts | No counts |
|-----------------------------------------------------------------------------------|------------|-----------|
| <b>SR Phase: Pre-Work</b>                                                         |            |           |
| Clearly Defined Research Question                                                 | 1,309      | 1,757     |
| Registered Protocol                                                               | 36         | 3,030     |
| Librarian or Database Expert Listed as Author                                     | 19         | 3,047     |
| Librarian or Database Expert Consulted                                            | 62         | 3,004     |
| Explicit Statement of Inclusion/Exclusion Criteria                                | 1,954      | 1,112     |
| <b>SR Phase: Searching</b>                                                        |            |           |
| Date Search Run                                                                   | 867        | 2,199     |
| Included a List of Sources Searched                                               | 2,937      | 129       |
| At Least One Full Database Search String Provided                                 | 708        | 2,358     |
| Database/s Search Strategy Provided (General Themes/Concepts)                     | 2,912      | 154       |
| Search Filters Identified                                                         | 1,271      | 1,795     |
| <b>SR Phases: Screening, Data Extraction &amp; Assessment</b>                     |            |           |
| PRISMA Flowchart or List of Excluded Studies with Reasons                         | 1,574      | 1,492     |
| Explicit Statement of Assessing for Bias of Included Studies                      | 368        | 2,698     |
| Explicit Statement of Multiple Reviewers or Explanation of Interrater Reliability | 464        | 2,602     |
| <b>SR Phase: Publishing</b>                                                       |            |           |
| SR and/or MA in Paper Title                                                       | 1,856      | 1,210     |

**TABLE 5.** Data extraction quality item paper counts by SR phase, broken down by if the article authors included SR and/or MA in the paper title (n=1,856), or not (n=1,210).

| Data extraction quality item                                                      | SR and/or MA in Paper Title |           | SR and/or MA not in Paper Title |           |
|-----------------------------------------------------------------------------------|-----------------------------|-----------|---------------------------------|-----------|
|                                                                                   | Yes counts                  | No counts | Yes counts                      | No counts |
| <b>SR Phase: Pre-Work</b>                                                         |                             |           |                                 |           |
| Clearly Defined Research Question                                                 | 859                         | 997       | 450                             | 760       |
| Registered Protocol                                                               | 32                          | 1,824     | 4                               | 1,206     |
| Librarian or Database Expert Listed as Author                                     | 17                          | 1,839     | 2                               | 1,208     |
| Librarian or Database Expert Consulted                                            | 52                          | 1,804     | 10                              | 1,200     |
| Explicit Statement of Inclusion/Exclusion Criteria                                | 1,328                       | 528       | 626                             | 584       |
| <b>SR Phase: Searching</b>                                                        |                             |           |                                 |           |
| Date Search Run                                                                   | 587                         | 1,269     | 280                             | 930       |
| Included a List of Sources Searched                                               | 1,781                       | 75        | 1,156                           | 54        |
| At Least One Full Database Search String Provided                                 | 467                         | 1,389     | 241                             | 969       |
| Database/s Search Strategy Provided (General Themes/Concepts)                     | 1,779                       | 77        | 1,133                           | 77        |
| Search Filters Identified                                                         | 818                         | 1,038     | 453                             | 757       |
| <b>SR Phases: Screening, Data Extraction &amp; Assessment</b>                     |                             |           |                                 |           |
| PRISMA Flowchart or List of Excluded Studies with Reasons                         | 1,123                       | 733       | 451                             | 759       |
| Explicit Statement of Assessing for Bias of Included Studies                      | 301                         | 1,555     | 67                              | 1,143     |
| Explicit Statement of Multiple Reviewers or Explanation of Interrater Reliability | 346                         | 1,510     | 118                             | 1,092     |

number produced in 2021. It is difficult to determine whether this most recent spike is directly related to the COVID-19 pandemic and the resultant 2020 lockdowns, but we can speculate that researchers unable to do more traditional scholarship may have taken the opportunity to produce review articles, with the resultant articles published the following year.

Another indicator of the breadth of SR growth is their presence across multiple document types and publishers. Table 2 shows a mix of journals and conferences, and, while we have not linked each of the 1,029 source titles to publishers or professional organizations, a small sampling suggests that SRs are being disseminated through most major publishers and associations. In our list of the top twenty sources, we can identify seven titles published by Elsevier, four from MDPI, three from IEEE, two from Taylor & Francis, two from ASEE, and one each from Emerald, Wiley, and ASCE. In terms of individual titles, Journal of Cleaner Production (an Elsevier journal, ISSN: 1879-1786) is by far the most prolific publisher of SRs. While we have no direct knowledge of their publishing strategies or practices, they

are an international journal that covers subject areas at the intersection of several disciplines that have embraced SR methods, notably industrial and environmental engineering. This would put them in a position to attract SRs from a wide range of different authors.

Looking at the geographic distribution of SR production (see Fig. 7) and librarian involvement (see Tables 6 and 7), the number of SRs produced in a given country does not appear to align with librarian involvement. While the United States did have the most articles reporting librarian consultation or co-authorship, Brazil and China were less well represented and other countries with lower overall SR numbers were better represented. The number of SRs with documented librarian involvement was low, overall. That said, this may be a cultural issue related to the roles and statuses of librarians and library professionals in different societies. Librarians who are in administrative or clerical staff roles, as opposed to academic collaborators, may have been more involved than has been reported.

As we can see from Fig. 5 and Fig. 6, not all engineering disciplines appear to have adopted SR methods at the same

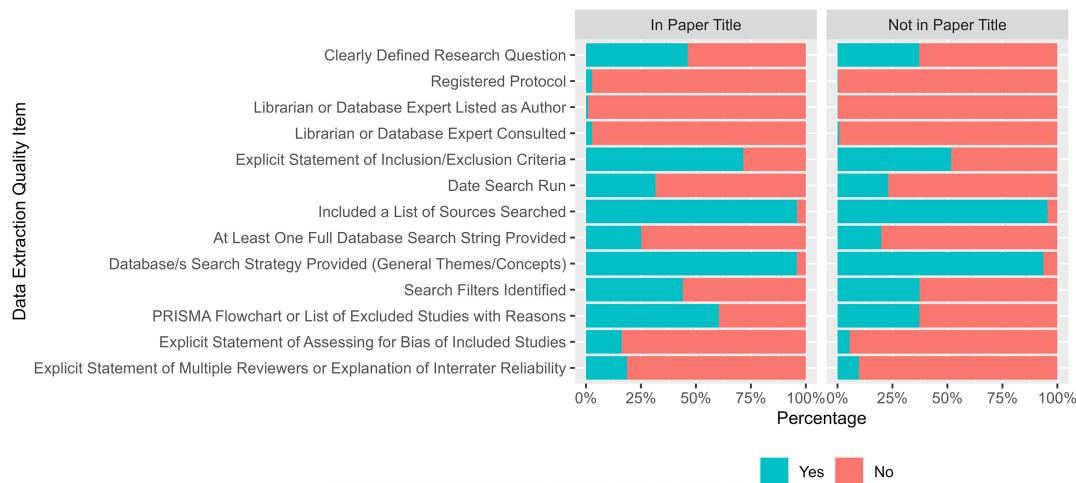


FIGURE 11. Percentage of SRs that adhered to (“yes”) or did not adhere to (“no”) each data extraction quality item, broken down by if the article authors included SR or MA in the paper title.

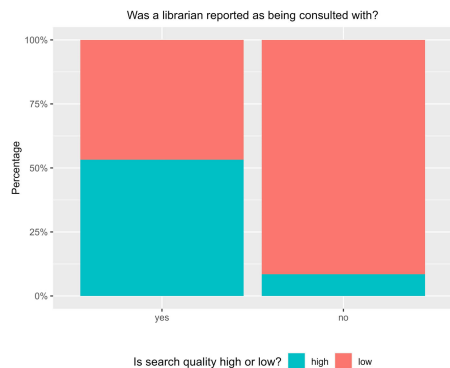


FIGURE 12. Percentage of SRs separated by if the study authors consulted with a librarian or database expert (n=62) or not (n=3,004), and by high or low search quality.

rate, or in proportion to the number of faculty in those disciplines. This could point to evidence-based practices entering engineering through associated disciplines. As an example, SRs are popular methodologies in business [38] and psychology [39], in education [35], [40], and especially in the health sciences [41]; this could explain the relatively high rate of SRs coming out of industrial engineering, engineering education, and biomedical engineering, respectively. With that said, more work needs to be done to explain the precise differences in how each engineering discipline uses these methods.

**B. RQ2: HOW CLOSELY ARE SYSTEMATIC REVIEWS THAT ARE PUBLISHED IN ENGINEERING FOLLOWING ESTABLISHED REPORTING GUIDELINES FOR THE METHODOLOGY?**

The intention behind the rigorous methods for SRs is three-fold: comprehensiveness, reproducibility, and transparency. As we investigate the included studies in more detail (see Table 4 and Fig. 10), we encounter a wide range of adherence to the reporting guidelines and methodological components required for SRs.

TABLE 6. Number of SRs by country that reported consulting with a librarian (n=62).

| Countries                                                                              | Number of SRs |
|----------------------------------------------------------------------------------------|---------------|
| United States                                                                          | 28            |
| Canada                                                                                 | 7             |
| China, Netherlands                                                                     | 4             |
| Australia, South Africa, United Kingdom                                                | 3             |
| Brazil, Denmark, Germany,Iran, Italy, Malaysia, Portugal, Saudi Arabia, Sweden, Turkey | 1             |

When documenting the search process, most studies perform well when it comes to reporting the baseline requirements of the general search strategy employed in obtaining the materials for their review (95%) and providing the sources searched (96%). However, there are issues associated with these. First, the quality of the documented search strategies is generally low (see Fig. 12). Poorly constructed search strategies lead to studies being missed and not included in analyses, which could have a major impact on the findings about a topic. Second, many authors listed only a database and not its associated platform (e.g., MEDLINE) or only reported a platform (e.g., Engineering Village) making it difficult for readers to ascertain the exact databases used. Third, the number of sources searched is typically small and often limited to searching only publications produced by a single publisher (e.g., ScienceDirect, Springerlink.) which further hinders the comprehensiveness. In addition, many studies do not report other information necessary for readers to replicate the search: only 41% included information about database filters, only 23% included at least one database search string, and only 28% included search date(s). Excluding such information removes any potential for reproducibility.

When studies document their screening processes, we see similar adherence issues that affect transparency and the potential for bias. For example, only a small majority of studies discuss their inclusion and exclusion criteria (64%) or include a detailed description of their screening process (51%). Far fewer include an explicit statement that multiple



**TABLE 7. Number of SRs by country with a librarian as a co-author (n=19).**

| Countries                                            | Number of SRs |
|------------------------------------------------------|---------------|
| United States                                        | 8             |
| Netherlands                                          | 3             |
| Canada                                               | 2             |
| Brazil, China, Italy, Portugal, Saudi Arabia, Turkey | 1             |

authors reviewed each article or an explanation of inter-rater reliability (15%), a key factor in reducing bias from the screening process itself. An even smaller group of studies (12%) perform what could be considered the hallmark of a systematic review: “critically appraises and reports the quality/risk of bias of the included studies.” We might expect to see better reporting of these components in the subset of studies that intentionally titled their articles a “systematic review” or “meta-analysis.” However, only slight improvements are observed (see Table 5 and Fig. 11). Without these key components, these studies are not serving the purpose of SRs to support evidence-based practice, but instead diluting it.

While it is difficult to surmise the reasons why these studies lacked most reporting components, it is clear that this is an issue across all areas of engineering. Engineering researchers are not harnessing the power of the SR methodology when they do not follow established guidelines for conducting and reporting. With this happening on a global scale, it is a cause of major concern for engineering. SRs are intended to be a way of finding out what the collective evidence says is effective or not regarding a particular topic. Poorly conducted SRs are no more helpful than traditional literature reviews that select studies to summarize and synthesize based on author preference. The fact that so many engineering researchers globally are interested in the SR methodology shows promise that future studies, if done well, could contribute significantly to the engineering literature, as has been the case with the medical community.

### C. RECOMMENDATIONS

There are two overarching and conflicting issues associated with SRs in engineering – the high growth rate across most engineering disciplines and geographic regions coupled with low methodological quality. To support and promote evidence-based practice in the disciplines of engineering, there is a need for education on three fronts: authors, publishers, and higher education administrators. The inconsistencies and deficiencies in reporting suggest authors’ overall lack of familiarity with the underlying principles and protocols associated with SRs. This hypothesis is further substantiated by the fact that only 1% of studies mention registering a protocol prior to initiating their study. These results also suggest that this lack of familiarity extends beyond the authors to the publishers given the sheer number of SRs published under the auspices of a very particular study type while not adhering to reporting requirements.

It is important that authors become conversant in the many types of literature reviews that exist, as an SR might not

suit the purpose of their particular study for many different reasons (e.g., number of co-authors, time restrictions). Grant and Booth’s article is an excellent primer for this purpose, as it lays out the details of fourteen different literature reviews and discusses the strengths and weaknesses of each [1]. If the authors decide to move forward with a SR, they should take time to identify and implement a set of established reporting guidelines such as PRISMA or conducting guidelines such as those proposed by Kitchenham and Charters [14]. These types of documents can serve as a roadmap for study design and help authors ensure the appropriate components are included. Something like the PRISMA Statement’s Checklist [42] is especially useful, as it offers a clear and concise path to follow and even offers an expanded version with more detail. It is also possible that collaboration with a librarian could help to educate engineering research teams to improve the overall quality of engineering SRs. There could be two levels of collaboration: librarian as a consultant focused on selecting the best sources and how to report a reproducible search or librarians as co-author focused on designing the search strategies and helping report the methods. This recommendation is supported by the low number of studies involving a librarian or database expert, either as a consultant (2) or as a co-author (0.6%).

Likewise, it is important that publishers supply clear guidance for SRs and ensure authors adhere to such guidance. This could be accomplished via the peer review process with a particular focus on the methods and results sections. Recruiting reviewers well-versed in the SR methodology or supplying reviewers with a checklist of methodological requirements could address such issues (PRISMA or equivalent could easily be repurposed for this purpose). Librarians could also serve in the capacity of special reporting requirement reviewers for these types of publications, examining only those portions of a publication. The Librarian Peer Reviewer Database (<https://sites.google.com/view/mlprdatabase/home>) is a potential source for locating librarians for this purpose and currently lists nearly 300 reviewers.

Finally, higher education administrators at research institutions with engineering programs need to make resources available to support the production of SRs. This could take the form of allocating library resources to establish a SR consultation service, funding new staff or faculty lines for search and methodology experts (librarians or otherwise), funding professional development opportunities to enhance SR expertise within the institution, developing programming to connect authors with the necessary information and expertise, etc. If the production of high-quality SRs is a disciplinary priority within engineering, commensurate support needs to be provided by the relevant institutions.

### D. LIMITATIONS

This scoping review has several limitations, as a result of our choices in developing our methods. Our search strategy (see Fig. 1 and Fig. 2) called out the terms “systematic

review,” “systematic literature review,” “meta-analysis” and “meta analysis” in specific metadata fields (title, abstract, keywords). If authors used different terms, we would have missed those articles. Alternatively, by excluding studies that used terms like “systematic review” but that did not report any data items we could extract, we may have inadvertently removed studies that did intend to use the systematic review methodology. Additionally, while we leveraged our expertise as subject specialist librarians to assign each article to an appropriate engineering discipline, we may not have adequately captured the multidisciplinary or interdisciplinary nature of some articles. Finally, the exclusion of articles not written in English could mean we are not able to see larger international trends.

### E. POTENTIAL FUTURE WORK

This data set, available in OSF (<https://osf.io/de3mq/>), presents several opportunities for future research, beyond the scope of our initial research questions. There may be the potential for additional research looking at our quality metrics with respect to specific disciplines and subdisciplines within engineering, similar to the engineering education subset study published in the Journal of Engineering Education [24]. Likewise, this data could be broken down by publisher or geography to assess protocol compliance across those populations. It could also be possible to extract additional information from this data set, such as specific SR protocols used and search reproducibility. However, the age of the data could become a concern, given the speed at which SR methods are being adopted in engineering disciplines. By the time additional data could be extracted from a data set of this size and complexity, it may be necessary to re-run and update the searches, requiring new rounds of screening and data extraction. This scoping review protocol could also be applied to disciplines outside of engineering, to investigate SR practices in those disciplines.

This study also raises questions that could be the subject of future research that looks to explain why these trends are occurring. We have quantified the growth of the SR in engineering disciplines and whether these reviews meet certain descriptive and quality criteria, but we can only offer supposition regarding causation. Can we directly attribute the spike in SR publications in 2021 to the COVID-19 pandemic? What is causing some engineering disciplines to adopt SR methodologies more quickly than others? Why are researchers claiming SR methods but not following the relevant protocols? Why are librarians and other search experts only minimally involved in engineering SR production, and what could or should be done to encourage researchers to seek out their expertise? Are libraries and librarians prepared and adequately resourced to support the volume of SRs being produced? How are engineering systematic reviews incorporating grey literature sources (i.e., non-traditional publications), such as government reports? What is the proper role of large language models in producing this type of study, and how are engineering researchers adapting to this new

technology? While we can provide suggestions for improving performance on quality-related criteria and supporting SR growth, there is also the opportunity for research that puts these suggestions into practice and assesses the results.

### V. CONCLUSION

This scoping review has shown that SRs are increasing in popularity across the broad field of engineering with 93% of the SRs included in this review being published since 2015. At the same time this review has found that the methods being reported in engineering SRs do not meet common reporting guidelines. The findings a SR produces are only as good as the data analyzed by the researchers and in the case of SRs this data comes from the database searches. For readers to critically assess if the results from a SR apply to their situation, they need to understand where the data comes from, just like any other research project. There is an opportunity for increasing education about different review methodologies within the engineering researcher community, facilitating collaborations with librarians to help establish and report the SR methods, encouraging higher education administrators at research institutions with engineering programs to provide funding and resources for SR support, and increasing awareness among journal editorial boards about best practices in SRs.

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